Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · FISHER'S EXACT

exact test for small categorical tables

Table 1. Contingency

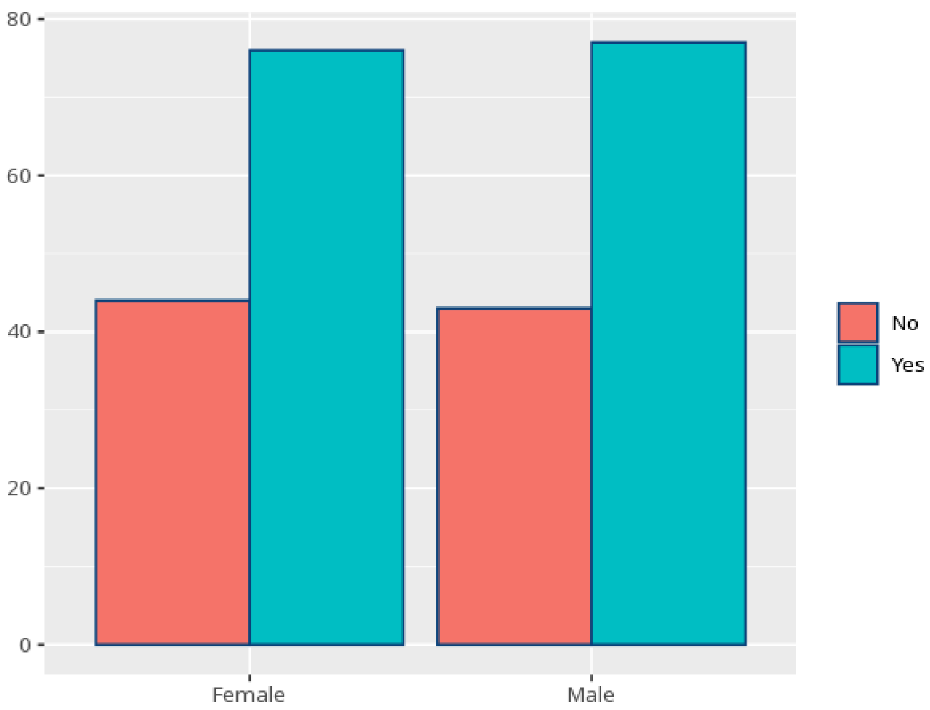
Row \ Column	No	Yes	Total
Female	44	76	120
Male	43	77	120
Total	87	153	240

Table 2. Fisher's exact test

p (exact)	Odds ratio (cond. MLE)	95% CI	Cramér's V [95% CI]
1.000	1.04	[0.59, 1.82]	—

the odds ratio is fisher.test()'s conditional MLE (not the sample cross-product) and, with its CI, is reported for 2×2 tables only; for larger tables the OR is undefined, so Cramér's V (effectsize::cramers_v) reports the strength of association alongside the exact p.

Figure. Cross-classification



Understanding the numbers

Row \ Column. The row-variable category heading each row of the contingency table. Here, the table has 2 row categories by 2 column categories.

Cell count. The observed count for that row/column combination (a plain count, unlike the expected/residual cells on the chi-square cards). Here, the table's cell counts total N = 240 across the table.

Cell count. The observed count for that row/column combination (a plain count, unlike the expected/residual cells on the chi-square cards). As with the first column, every cell here is a plain count, across 2 column categories.

Total. The row or column margin - the sum across that row or column. Here, the table's totals sum to N = 240.

p (exact). The exact probability of a table this extreme (or more extreme) if the two categories were truly unrelated. Here, p (exact) = 1.000 - at or above your significance threshold, no significant association detected.

Odds ratio. For 2×2 tables, how many times more likely the outcome is in the first row/group vs. the second - this is fisher.test()'s conditional MLE, not the plain sample cross-product. Here, OR = 1.04 - the outcome is more likely in the first row/group (check the row/column order before reading direction).

95% CI (odds ratio). The confidence interval around the odds ratio, reported alongside it for 2×2 tables. Here, 95% CI = [0.59, 1.82].

How to read this test

An exact alternative to chi-square for small samples. A p below alpha means the two categories are associated; for 2×2 tables the odds ratio quantifies the association — this is fisher.test()'s conditional MLE (the non-central hypergeometric estimate), not the sample cross-product. OR>1 means the outcome is more likely in the first row/group, OR<1 less likely, OR=1 no association; check the row/column order before reading direction, since the OR reflects the table's orientation. For tables larger than 2×2 the odds ratio is undefined, so Cramér's V (0–1) reports the strength of association instead. Your significance threshold (α) is 0.05.

APA template: A Fisher's exact test of gender by passed_course gave p = 1.000 (OR=1.04, 95% CI [0.59, 1.82]).

Statistical basis: Fisher's exact test (conditional MLE odds ratio for 2x2 tables) (Fisher, R. A. (1922). "On the interpretation of χ^2 from contingency tables, and the calculation of P." Journal of the Royal Statistical Society, 85(1), 87-94.); Cramér's V effect size (larger-than-2×2 tables) (Ben-Shachar MS, Lüdecke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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